IRIS FLOWER CLUSTERING PROJECT

Week 3 Mini Project | Machine Learning & AI

Project Area	Unsupervised Machine Learning
Algorithm	K-Means Clustering (K = 3)
Dataset	Iris Dataset (UCI / Kaggle)
Key Tasks	Cluster visualization, PCA, and comparison with true labels

1. Introduction

Clustering is an unsupervised machine-learning technique used to group data points based on similarity. In this project, the Iris dataset is grouped into three clusters using the K-Means algorithm. The resulting clusters are visualized and compared with the dataset's known species labels.

2. Objectives

• Apply K-Means clustering to the Iris dataset with K = 3.

• Visualize the generated clusters using a scatter plot.

• Use Principal Component Analysis (PCA) to reduce the feature dimensions for visualization.

• Compare predicted cluster assignments with the true Iris species labels.

• Understand feature scaling and basic model saving/loading concepts.

3. Dataset Description

The Iris dataset contains measurements of iris flowers, commonly including sepal length, sepal width, petal length, and petal width. Each row represents one flower. The dataset also provides species labels, which are used only for comparison after clustering, not to train K-Means.

4. Methodology

• Load the Iris dataset and separate the four numerical measurements from the species labels.

• Standardize the features so that measurements on different scales contribute comparably.

• Fit K-Means with three clusters.

• Use PCA to project the four features into two dimensions for a scatter-plot view.

• Compare the cluster assignments with the true labels; cluster numbers themselves do not directly correspond to species names.

5. Python Implementation

```
# Install if needed:
# pip install scikit-learn pandas matplotlib

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import pandas as pd

# Load data
iris = load_iris()
X = iris.data
true_labels = iris.target

# Scale features
X_scaled = StandardScaler().fit_transform(X)

# K-Means clustering (K = 3)
model = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = model.fit_predict(X_scaled)

# Reduce dimensions with PCA
X_pca = PCA(n_components=2).fit_transform(X_scaled)

# Visualize predicted clusters
plt.figure(figsize=(8, 5))
plt.scatter(X_pca[:, 0], X_pca[:, 1],
c=clusters, cmap="viridis", s=45)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Iris Flower Clusters (K-Means)")
plt.tight_layout()
plt.show()

# Compare cluster IDs with true species labels
comparison = pd.DataFrame({
"True species": [iris.target_names[i] for i in true_labels],
"Predicted cluster": clusters
})
print(comparison.head(10))

# Optional: save and load the model
import joblib
joblib.dump(model, "iris_kmeans.joblib")
loaded_model = joblib.load("iris_kmeans.joblib")
```

6. PCA and Cluster Visualization

PCA (Principal Component Analysis) transforms the original features into a smaller set of components that preserve as much variation in the data as possible. Here, two principal components are used to plot the four-dimensional flower measurements on a two-dimensional chart. The plot shows the K-Means cluster assignments; the exact appearance depends on the data and preprocessing.

7. Comparing Clusters with True Labels

K-Means is unsupervised, so it does not use the species labels when forming clusters. After fitting, compare each predicted cluster with the known species labels to inspect how well the groupings align. Cluster IDs (0, 1, and 2) are arbitrary identifiers and should not be interpreted as species names without an additional matching step.

8. Saving and Loading the Model

The example uses joblib to save the fitted K-Means model to **iris_kmeans.joblib** and load it later. Keep the same preprocessing steps when applying the model to new flower measurements.

9. Conclusion

This mini project demonstrates unsupervised learning through K-Means clustering on the Iris dataset. Feature scaling helps make the measurements comparable, while PCA supports two-dimensional visualization. Comparing predicted clusters with the true species labels provides a practical way to review the clustering result.

Expected skill gain: Unsupervised machine learning, clustering, dimensionality reduction, and visualization.